

06_Correlation Analysis

Correlation Analysis in machine learning (ML) is a statistical method used to measure the strength and direction of the relationship between two or more variables. It's a crucial step in **exploratory data analysis (EDA)** to understand how features in your dataset are related.

Why is Correlation Analysis Important in ML?

- **Feature Selection:** Helps identify which features are strongly related to the target variable.
- **Multicollinearity Detection:** Helps detect when two or more independent variables are highly correlated, which can affect model performance, especially in linear models.
- **Insights & Understanding:** Gives a clearer picture of relationships in your data.

Common Correlation Metrics

Metric	Best For	Range	Notes
Pearson	Linear relationships	-1 to 1	Most common; assumes normality
Spearman	Monotonic relationships	-1 to 1	Uses rank-order; non-parametric
Kendall Tau	Ordinal data	-1 to 1	More robust for small datasets

Coefficient (r)

1.0

0.7 to 0.9

0.4 to 0.6

0.1 to 0.3

0

-0.1 to -0.3

-0.4 to -0.6

-0.7 to -0.9

-1.0

Interpretation

Perfect positive correlation

Strong positive

Moderate positive

Weak positive

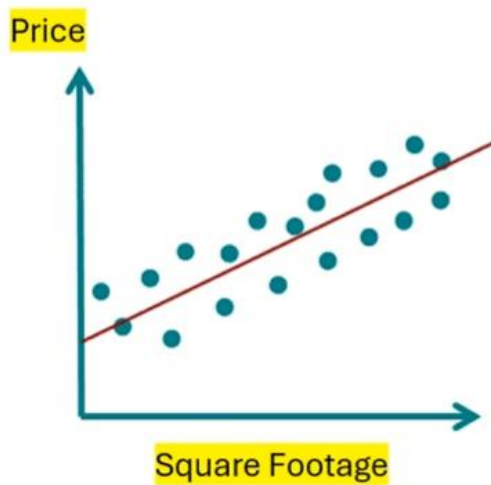
No correlation

Weak negative

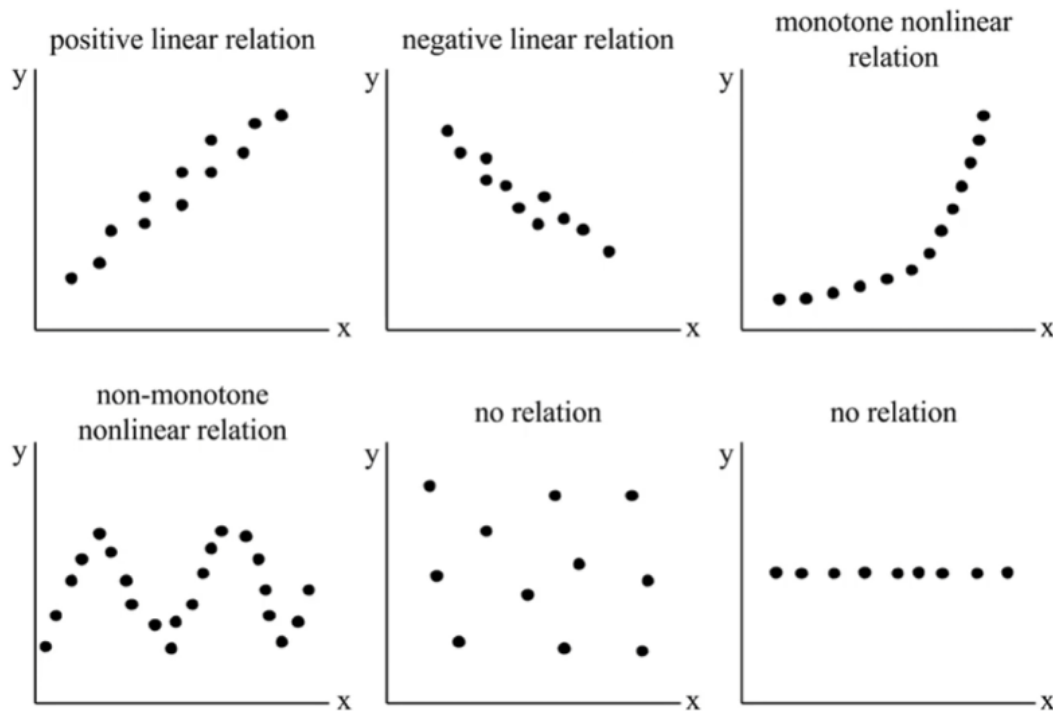
Moderate negative

Strong negative

Perfect negative correlation



Correlation in regression:



Pearson Correlation Coefficient:

Pearson Correlation Coefficient: Correlation is always between two variables at a time. When we show a correlation matrix, it just lists all the pairwise correlations between every **pair of variables**.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \cdot \sqrt{\sum (y_i - \bar{y})^2}}$$

Where:

- x_i, y_i are individual values of the two variables
- $\bar{x}, \bar{y}$ are their means
- r is the **correlation coefficient** (between -1 and 1)

Dataset			
Sample	Size (x)	Price (y)	Age (z)
1	1400	250000	5
2	1600	300000	10
3	1800	350000	15

Given data

Sample	Size (X)	Price (Y)	Age (Z)
1	1400	250000	5
2	1600	300000	10
3	1800	350000	15

We want:

$$r_{XY} = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2} \cdot \sqrt{\sum(Y_i - \bar{Y})^2}}$$

- X_i, Y_i are the data points
- $\bar{X}, \bar{Y}$ are the means of X and Y
- r_{XY} ranges from -1 to 1:
 - 1 → perfect positive correlation
 - -1 → perfect negative correlation
 - 0 → no linear correlation

Step 1: Compute the means

$$\bar{X} = \frac{1400 + 1600 + 1800}{3} = \frac{4800}{3} = 1600$$

$$\bar{Y} = \frac{250000 + 300000 + 350000}{3} = \frac{900000}{3} = 300000$$

Step 2: Compute deviations and their products

X	Y	X - $\bar{X}$	Y - $\bar{Y}$	(X - $\bar{X}$)(Y - $\bar{Y}$)
1400	250000	-200	-50000	200 * 50000? Wait carefully!

Let's calculate carefully:

- $X - \bar{X} = 1400 - 1600 = -200$
- $Y - \bar{Y} = 250000 - 300000 = -50000$
- Product: $(-200) * (-50000) = 10,000,000$

Next row:

- $X - \bar{X} = 1600 - 1600 = 0$
- $Y - \bar{Y} = 300000 - 300000 = 0$
- Product = 0

Next row:

- $X - \bar{X} = 1800 - 1600 = 200$
- $Y - \bar{Y} = 350000 - 300000 = 50000$
- Product = $200 * 50000 = 10,000,000$

$$\sum(X_i - \bar{X})(Y_i - \bar{Y}) = 10,000,000 + 0 + 10,000,000 = 20,000,000$$

Step 3: Compute squared deviations

$$\sum(X_i - \bar{X})^2:$$

$$(-200)^2 + 0^2 + 200^2 = 40000 + 0 + 40000 = 80000$$

$$\sum(Y_i - \bar{Y})^2:$$

$$\begin{aligned} (-50000)^2 + 0^2 + 50000^2 &= 2,500,000,000 + 0 + 2,500,000,000 \\ &= 5,000,000,000 \end{aligned}$$

Step 4: Compute r_{XY}

$$r_{XY} = \frac{20,000,000}{\sqrt{80,000} \cdot \sqrt{5,000,000,000}}$$

Compute the square roots carefully:

$$\sqrt{80,000} \approx 282.84$$

$$\sqrt{5,000,000,000} \approx 70,710.68$$

Multiply denominators:

$$282.84 * 70,710.68 \approx 20,000,000 \text{ (very close)}$$

$$r_{XY} = \frac{20,000,000}{20,000,000} = 1$$

✅ Perfect positive correlation between **Size** and **Price**.

Step 5 (Optional): Correlation with Age

We can do r_{YZ} or r_{XZ} in the same way. Let's check **Price (Y) vs Age (Z)** quickly:

$$\bar{Z} = \frac{5 + 10 + 15}{3} = 10$$

Compute deviations:

Z	Y	Z - $\bar{Z}$	Y - $\bar{Y}$	Product
5	250000	-5	-50000	250,000
10	300000	0	0	0
15	350000	5	50000	250,000

$$\Sigma(Z - \bar{Z})(Y - \bar{Y}) = 500,000$$

$$\Sigma(Z - \bar{Z})^2 = (-5)^2 + 0^2 + 5^2 = 50$$

$$\Sigma(Y - \bar{Y})^2 = 5,000,000,000$$

$$\sqrt{50} \approx 7.071, \sqrt{5,000,000,000} \approx 70,710.68$$

$$\text{Denominator} = 7.071 * 70,710.68 \approx 500,000$$

$$\Rightarrow r_{YZ} = 500,000 / 500,000 = 1$$

Conclusion: In this dataset, all variables are perfectly positively correlated (linear growth).

An example correlation table with **4 variables** commonly found in a housing dataset, showing how they might relate to House Price:

Feature	House Price	Size (sqft)	Number of Rooms	Age of House
House Price	1.00	0.88	0.75	-0.65
Size (sqft)	0.88	1.00	0.70	-0.40
Number of Rooms	0.75	0.70	1.00	-0.30
Age of House	-0.65	-0.40	-0.30	1.00

Interpretation: House Price has-

- a **strong positive** correlation with **Size (sqft)** (0.88)
- a **moderate positive** correlation with **Number of Rooms** (0.75)
- a **strong negative** correlation with **Age of House** (-0.65), meaning older homes tend to